

PolicySim

Does a Cash Transfer Policy Actually Work?

A statistical evaluation of a generative agent-based economic simulation
MSc Economics portfolio project — results report

1. What this is

PolicySim is a small agent-based economy (households, firms, a government) that reasons in plain language each round and reacts to policy shocks a user can trigger live. This report focuses on one question the project set out to answer with real statistics rather than narrated vibes: **does a flat cash transfer to low/mid-income households actually change the simulated economy, and if so, how?**

All numbers below are computed live, at report-generation time, by importing and calling the actual experiment functions in `experiments/` -- nothing in this PDF is a hand-typed number. Every experiment can also be run standalone with `python -m experiments.<name>`.

1.1 Method

40 independent simulation runs per condition (different random seed each run), 20 rounds per run, `cash_transfer` applied from round 1 in the treatment group and never in the control group. Final-round outcomes are compared with Welch's t-test (unequal variances) and Cohen's d for effect size -- except where a metric shows ~zero within-group variance in both conditions, in which case an exact deterministic difference is reported instead of a t-test (see §1.2).

1.2 A note on 'degenerate' metrics

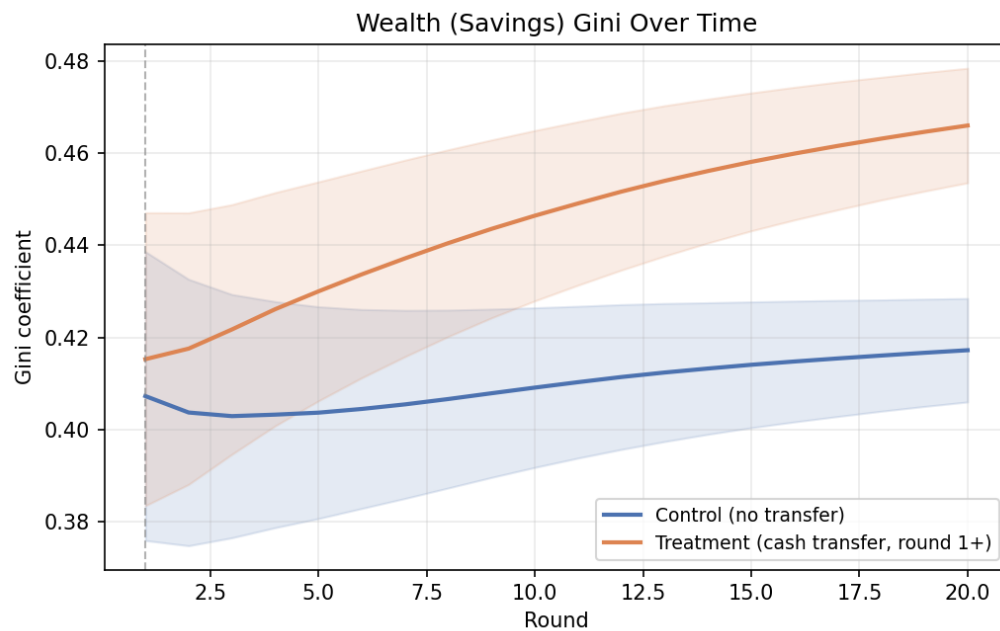
Some metrics in this model are fully deterministic given tier/employment/policy -- e.g. disposable income has no random input at all when every household is employed every round, so its Gini coefficient comes out *identical* across all 40 seeds within a condition. A significance test comparing two zero-variance samples is mathematically degenerate (division by ~0), not just "very significant" -- earlier drafts of this analysis reported such cases as e.g. Cohen's d = 0.000 sitting next to $p < 0.0001$, which is internally inconsistent and was a real bug, not a stylistic choice. Tables below mark these cases "exact" and report the true deterministic difference instead.

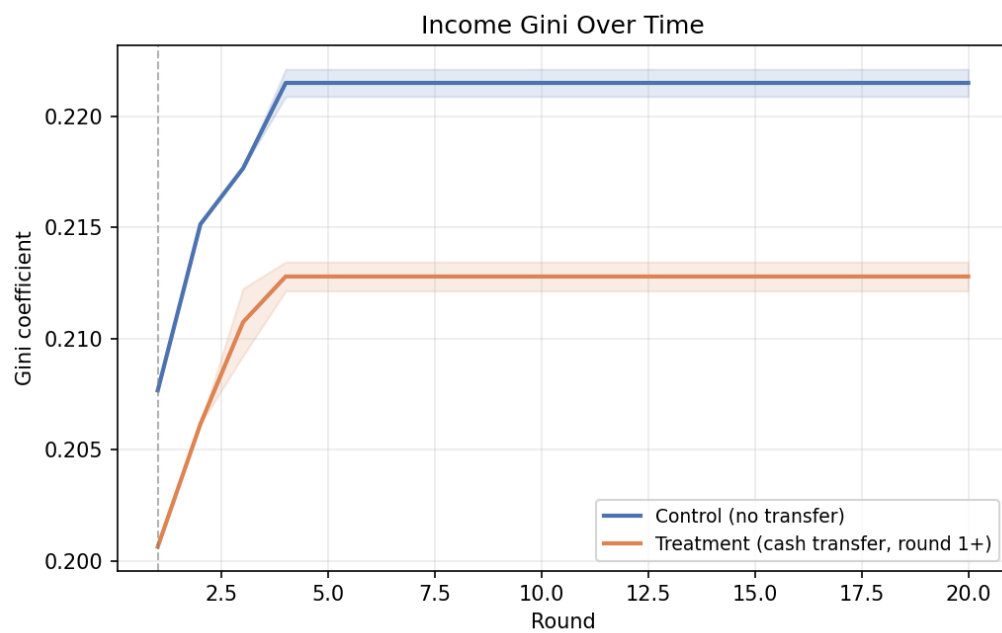
2. Headline result: wealth inequality up, income inequality down

The core, counter-intuitive finding: a cash transfer that directly boosts low/mid-income households' *income* makes income inequality better, but makes wealth (savings) inequality *worse*.

Metric	Control mean	Treatment mean	t-stat	p-value	Cohen's d
Wealth Gini	0.4172	0.4661	18.175	<0.0001	4.06
Income Gini	0.2215	0.2128	exact	diff=-0.0087	n/a
Price level	1.0232	1.0249	0.578	0.565	0.13
Total spending	358042.3125	406896.6394	25.354	<0.0001	5.67
Unemployment	0.3344	0.3219	-3.961	0.000	-0.89

Why: the transfer is a fixed rupee amount, small relative to high-tier households' base income. High earners keep compounding savings at their usual pace. Low/mid households spend a larger share of their (now slightly bigger) income rather than saving it, so nominal wealth inequality widens even though the transfer helps low/mid *income* directly — visible in income Gini falling in the same runs.





Mean $\pm$ 1 std across 40 runs per condition, round by round. The two Gini measures diverge almost immediately after the transfer starts and stay apart for the full 20 rounds — this is a stable trend, not a single lucky snapshot.

3. Does a bigger transfer make wealth inequality worse?

Not linearly. Averaging 15 seeds per magnitude removes enough noise to see the actual shape: wealth Gini jumps as soon as *any* transfer is applied, then slightly recedes as the transfer keeps growing — a threshold effect with a mild ceiling, not a dial that keeps making things worse the harder you push it.

Magnitude	Wealth Gini	Income Gini	Price level
0 (none)	0.4191	0.2216	1.0226
250	0.4601	0.2195	1.0226
500	0.4593	0.2173	1.0226
1000 (default)	0.4576	0.2132	1.0226
2000	0.4542	0.2052	1.0226
4000	0.4475	0.1906	1.0331

Income Gini, in contrast, falls smoothly and monotonically as the transfer grows (0.2216 → 0.1906) — exactly what you'd expect from a transfer targeted at low/mid earners. Wealth Gini jumps from 0.4191 (no transfer) to 0.4601 the moment *any* transfer is applied (magnitude=250), peaks at magnitude=250 (0.4601), then drifts back down to 0.4475 at magnitude=4000. **Takeaway:** don't read the wealth-Gini result as “bigger transfer = worse for wealth inequality”; it's closer to “any transfer at all widens wealth inequality at this population size, with a ceiling on how much.”

4. Unemployment: real in the default population, and it responds to policy

The default population (24 households, 4 firms offering 18 starting jobs) has genuine labor-market slack (~33% unemployment at baseline) -- this used to require a special override to demonstrate (an earlier, smaller default population left zero slack, a real bug documented in §6.3 of the README). With real slack, the cash transfer's demand boost measurably moves employment on its own, no special scenario needed:

Unemployment falls significantly with the transfer ($p=0.0002$, 33.4% → 32.2%, Cohen's $d=-0.89$) — see the headline table in §2 for the full metric set on this same run.

As a separate robustness check in the opposite direction: does the wealth-Gini finding (§2) still hold if we deliberately remove all labor-market slack, so unemployment is structurally pinned at 0% (30 jobs vs. 24 households, no room for the transfer to affect employment at all)? If the wealth-Gini effect disappeared here, that would mean employment risk -- not differing marginal propensity to consume -- was secretly driving the whole result.

Metric	Control mean	Treatment mean	t-stat	p-value	Cohen's d
Unemployment rate	0.0000	0.0000	exact	diff=+0.0000	n/a
Wealth Gini	0.5179	0.5642	34.444	<0.0001	7.70

Income Gini	0.3094	0.2994	exact	diff=-0.0100	n/a
Price level	0.9785	1.0041	8.149	<0.0001	1.82

Wealth Gini still rises significantly ($p < 0.0001$, Cohen's $d = 7.70$) even with zero unemployment risk possible — confirming the mechanism really is differing marginal propensity to consume by tier (\$6), not employment risk in disguise.

5. Robustness check: does the headline result survive a different population size?

The wealth-Gini finding was re-tested with double the household count (48 households instead of 24, same firms, same design) to check it isn't an artifact of a specific small population.

Metric	Control mean	Treatment mean	p-value	Cohen's d
Wealth Gini (baseline, n=24 hh)	0.4172	0.4661	<0.0001	4.06
Wealth Gini (2x households, n=48 hh)	0.4112	0.4484	<0.0001	3.67

The direction and significance of the effect held under 2x population (effect size similar: $d = 3.67$ vs. $d = 4.06$). Absolute Gini levels differ because a larger, evenly-tiered population changes the wealth distribution's baseline shape, but the treatment-vs-control gap is the same story: this is a stable emergent property of the mechanism, not a fluke of one specific population size.

6. Why this matches real economics

The mechanism behind the divergence — poorer households spend a larger share of additional income while richer households save more of theirs — mirrors well-documented differences in marginal propensity to consume (MPC) by income/wealth level. PSID-based estimates put MPC around 0.15 for the lowest wealth quintile vs. roughly 0.06 for the highest (Fisher, Johnson, Smeeding & Thompson); recession-calibrated estimates from the Penn Wharton Budget Model show a similar gradient by income quintile (roughly 0.55 down to 0.12). PolicySim's heuristic household brain is loosely tuned to that gradient (see `brain.py` and `simulation.py`), so the wealth-Gini-up / income-Gini-down split isn't an arbitrary quirk of this toy model — it's the same distributional mechanic that shows up in real transfer-payment research, reproduced here in miniature.

7. Limitations

- **Population size.** 10–20 households and 2 firms is a toy economy. Directional results are consistent under a 2x robustness check, but this is not a claim about real-world magnitudes.

- **Heuristic reasoning only.** All results here use the zero-dependency heuristic brain, not an LLM backend (ollama/groq); those backends introduce their own variance not characterized here.
 - **Single policy studied in depth.** The subsidy cut, minimum wage, and luxury tax policies exist and are unit-tested, but this report focuses statistical depth on cash_transfer specifically.
 - **20-round horizon.** Longer-horizon dynamics (e.g. compounding wealth effects over 100+ rounds) are untested here.
 - **Multiple comparisons.** Five metrics are reported per experiment; treat each p-value as part of a larger comparison set rather than a single pre-registered test.
 - **Degenerate-metric detection uses a fixed variance floor (1e-6).** Calibrated against this project's actual metric scales (see §1.2 and experiments/cash_transfer_effect.py), not derived analytically -- reasonable here, but wouldn't necessarily transfer to a differently-scaled model unmodified.
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8. Reproducing these results

```
python -m experiments.cash_transfer_effect    # §2 and §5
python -m experiments.magnitude_sweep_cash_transfer    # §3
python -m experiments.unemployment_slack    # §4
python -m experiments.generate_charts    # regenerates the charts above
python build_report.py    # regenerates this entire PDF, live, from scratch
pytest tests/ -v    # regression suite
```

Generated from the PolicySim repository by running the actual experiment code -- every number above was computed at build time, not hand-typed. See README.md for the full project writeup, the policy catalogue, and the live-demo trigger system.